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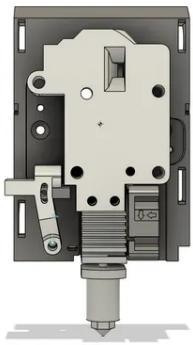
3D MODEL ONLY



QIDI X-Max 3 (Post Extruder) Filament cutter (Updated)

AskewedBox

VIEW IN BROWSER

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Summary

QIDI X-Max 3 extruder with ECAS fitting and runout sensor and extruder disengage slot.

3D Printers > Other Printer Parts & Upgrades

Tags: extruder filamentcutter qidi qidixmax3

I believe I have a working Beta

Updated: (4/22/2025)

https://vimeo.com/manage/videos/1077693101/502d7b66e4

This is a work in progress! Proceed with caution.

Additionally, this will also require a new homing procedure.

The goal of this project is to add a sturdy and integrated filament cutter that minimizes waste while keeping the rest of the printer as stock as possible.

This design is a modified version of the original extruder body. It includes a filament cutter "blade port" positioned directly above the hotend and has been simplified for easier printing.

Side note: QIDI originally had a design for this but not on the release model I have.

In addition to the filament cutter, this modification incorporates a filament runout sensor placed above the extruder. The runout sensor utilizes a 5 mm bearing (original this was a 4.5mm BB) and an Omron D2F style limit switch: https://www.digikey.com/en/products/detail/omron-electronics-inc-emc-div/D2F/83270

Required:

Printed Parts

- Extruder Cover (Front (no supports needed), Left Panel, Right Panel)
- Lever Arm
- Lever Arm Spacer
- Blade Holder (print a few)
    - Notes
        - I printed in ABS with 40% infill and 4 walls.
        - The Left and Right Panels as well as the lever arm need support everything else should be fine.
        - I was having issues with the tabs snapping no matter how I printed them so I broke it up into multiple parts them printed for strength. As a plus this makes it easier to transfer the extruder fan.

Filament Sensor

- 1x 5mm Bearing
- 1x Omron D2F style limit switch
- 2x Self-tapping M2x8mm screws
- Optional but helpful: long M2 screws to adjust the limit switch

Extruder

- ECAS04
- 4x M3 heat set inserts of any length for the extruder body
- 2x M3x4 heat set inserts for the hotend mount holes
- 2x M3 heat set inserts of any length for the cover if you print this in multiple pieces.

Cutter

- 18mm Snap-off Blade with 8mm segments - I bought mine at Harbor Freight.
- 1x M3X20 screw
- 4 mm OD, 0.5mm Wire Size, 15 mm compression spring - It is important that these are stiff, I got mine from my Box Turtle Kit
    - These should work but are **untested** https://www.amazon.com/gp/product/B0C33BYX83/
- Superglue

Instructions (in work):

Extruder

- Insert the ECAS fitting (this should be hard to insert, I pressed it in using a vise)
- Filament Sensor
    - Insert the 5mm bearing ensuring that it moves freely.
    - Insert the limit switch in the correct orientation
    - Insert the optional M2 screws to apply pressure light pressure to the switch.
    - To adjust the placement of the switch take a piece of filament and insert it in the extruder listening for a click as you insert it.
        - If you don't hear a click tighten the M2 screws and try again.
        - It is critical that you ensure that the switch is triggered by the filament before setting its placement permanently.
    - Use the Self-tapping M2 screws set the switch location.
- Transfer the extruder hardware from the stock extruder. Be careful not to lose the tension spring or the gear bearings. The print is designed without tolerances for the bearings or gear pins, so some hardware mainly the dowel pins, may be difficult to insert.
    - I pressed in the bearing and reamed the dowel hole with a 3mm bit.
        - This will still be tight fit but easer to remove if needed.
- Insert 4 M3 heat-set inserts into the extruder body flush.
- For the hotend, ensure the inserts are 4mm or less.
- Reassemble the extruders gears and reattach the back plate
- Check that the gears turn freely
- **Notes**
    - Don't attach the hotend yet; you will want it off to verify blade length and smooth movement.
    - Ensure that no plastic is displaced that will cause the blade to bind.

        - The have been increased to 4.2 mm but might still be slightly undersized
            - Ensure the tenson spring is not binding on the plastic was displaced by the insert.

Blade Holder

-----Use Caution when handling and snaping the blades!

- The holder is designed for a 8mm wide section from a snap-off box cutter blade
    - The blade needs to be snapped to a length of of approximately ~18mm (measures from the cutting edge) or slightly less.
    - Insert the blade into the holder such that the lever assembly is around 25.7mm (This is what worked for me through guess and check)
    - Glue the blade into the holder.
    - Now is a good time to check that the blade is free moving in the extruder blade slot and that it extends past the filament hole by at least 1mm.
    - **Note**
        - I used a vise and pliers to snap this to make clean(er) breaks but even then it wasn't consistent.
        - The goal is to have a blade length that is long enough to cut the filament but isn't so long that it binds when retracted by the spring.
        - It is hard to snap this exactly and this might take a few tries.
        - This section will probably require trial and error.
            - The lever arm length and hole depth were fiddly to get just right and may require adjusting to get the correct blade length.

Lever Arm

- Screw the assembled holder into the lever with two short M2 screws,
    - Don't over tighten these since they should be free moving.
- Insert the M3x20 mm screw into the lever arm (these should be relatively free moving on the screw.
- Insert the screw into the printed spacer.
- Screw the lever are onto the preexisting hole with a threaded insert next to the hotend

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Untested Best Guess at Hotend Values BoxTurtle MMU:

tool_stn: 44.5
tool_stn_unload: 96.6
variable_retract_length: 23.7
variable_pushback_length: .9 **(most likely wrong)**

This remix is based on

GitHub - vega-d/qidi_mods: a set of mods to qidi printers. Assorted by model.

Model files

V1 3 files

head_cutter.3mf

blade-holder.step

qidi-toolhead-cutter-v1.step

V0 2 files

qidi_extruder_cutter.step

qidi_extruder_assembly.step

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